

Cyber Experts - Group Project Proposal

**A Lightweight Machine Learning-Based Network Monitoring and Intrusion
Detection System**

Introduction

Modern computer networks face a growing number of cyber threats, including unauthorized access attempts, network scanning, and denial-of-service attacks. Traditional intrusion detection systems often rely on predefined signatures and rule-based mechanisms, which may struggle to identify previously unseen attack patterns. As network traffic continues to increase in complexity, there is a need for intelligent systems capable of analyzing traffic behavior and identifying anomalies automatically.

This project proposes the development of a lightweight network monitoring and intrusion detection system that leverages machine learning techniques to identify suspicious network activities. The system is designed for small organizations, educational laboratories, and research environments where a simple, cost-effective security monitoring solution is required.

The proposed solution will collect network traffic information, process selected traffic features, classify traffic as normal or suspicious using a machine learning model, and present the results through a web-based dashboard.

Problem Statement

Many organizations depend on traditional monitoring tools that require manual configuration and frequent signature updates. Such systems may fail to detect emerging threats or abnormal traffic behavior that does not match known attack signatures.

The challenge is to develop a system that can automatically learn traffic patterns from historical data and assist network administrators in identifying potentially malicious activities with minimal manual intervention.

Project Objectives

Main Objective

To design and implement a machine learning-based network monitoring system capable of detecting suspicious network traffic and presenting the results through a simple monitoring dashboard.

Specific Objectives

- Capture and analyze network traffic data.
- Extract essential network traffic features.
- Train a machine learning model using publicly available intrusion detection datasets.
- Classify traffic as normal or suspicious.
- Store detection logs in a PostgreSQL database.
- Visualize monitoring results through a web interface.

Scope of the Project

Included Features

- Basic network traffic monitoring.
- Traffic feature extraction.
- Machine learning-based anomaly detection.
- Detection log storage.
- Dashboard for viewing traffic statistics and alerts.

Proposed Methodology

Phase 1: Dataset Collection

Network traffic datasets such as NSL-KDD and CICIDS2017 will be studied and selected for model training.

Phase 2: Data Preprocessing

The collected dataset will undergo:

- Data cleaning
- Feature selection
- Label encoding
- Data normalization

Phase 3: Model Development

A Random Forest classifier will be trained to distinguish between normal and suspicious traffic. The model will be evaluated using standard performance metrics.

Phase 4: Backend Development

A Node.js and TypeScript backend will:

- Receive network traffic information.
- Communicate with the machine learning prediction service.
- Store prediction results in PostgreSQL.

Phase 5: Frontend Development

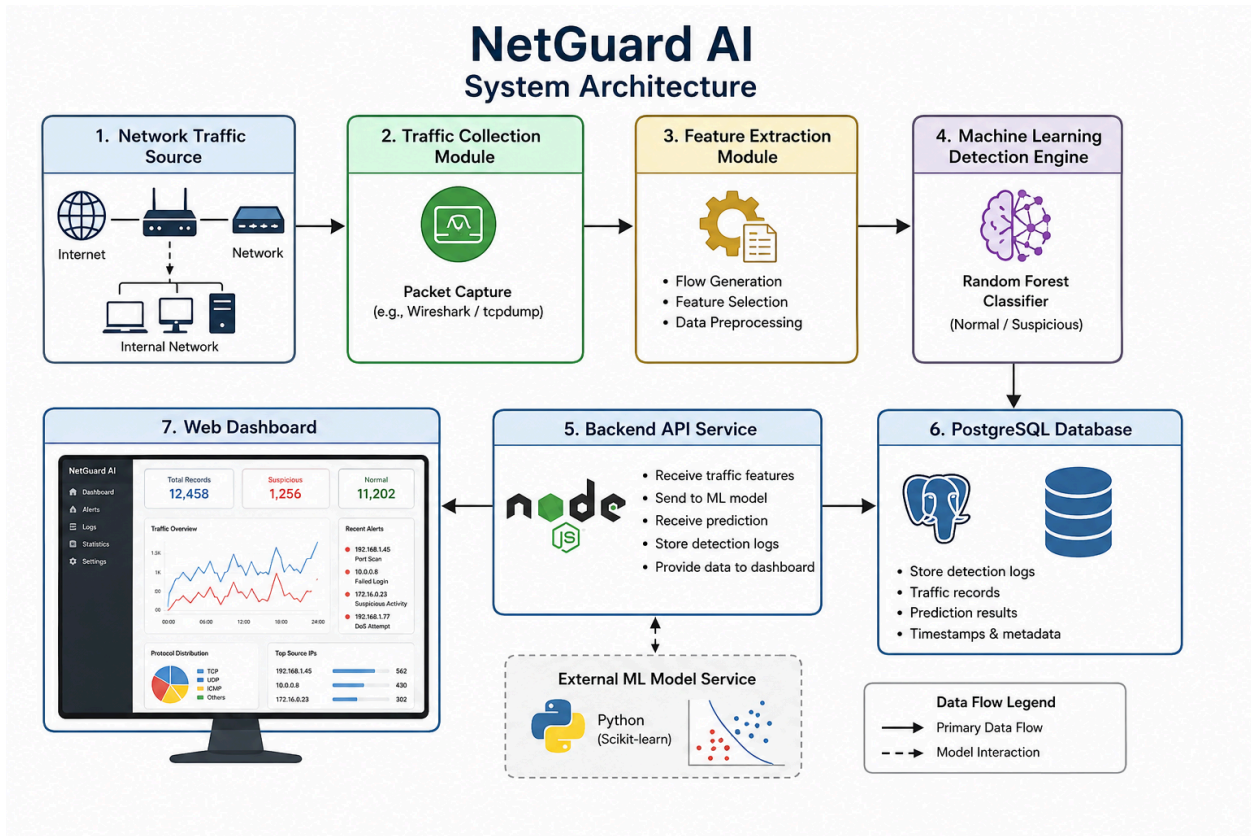
A React-based dashboard will display:

- Total monitored traffic records.
- Number of suspicious detections.
- Number of normal detections.
- Recent detection logs.

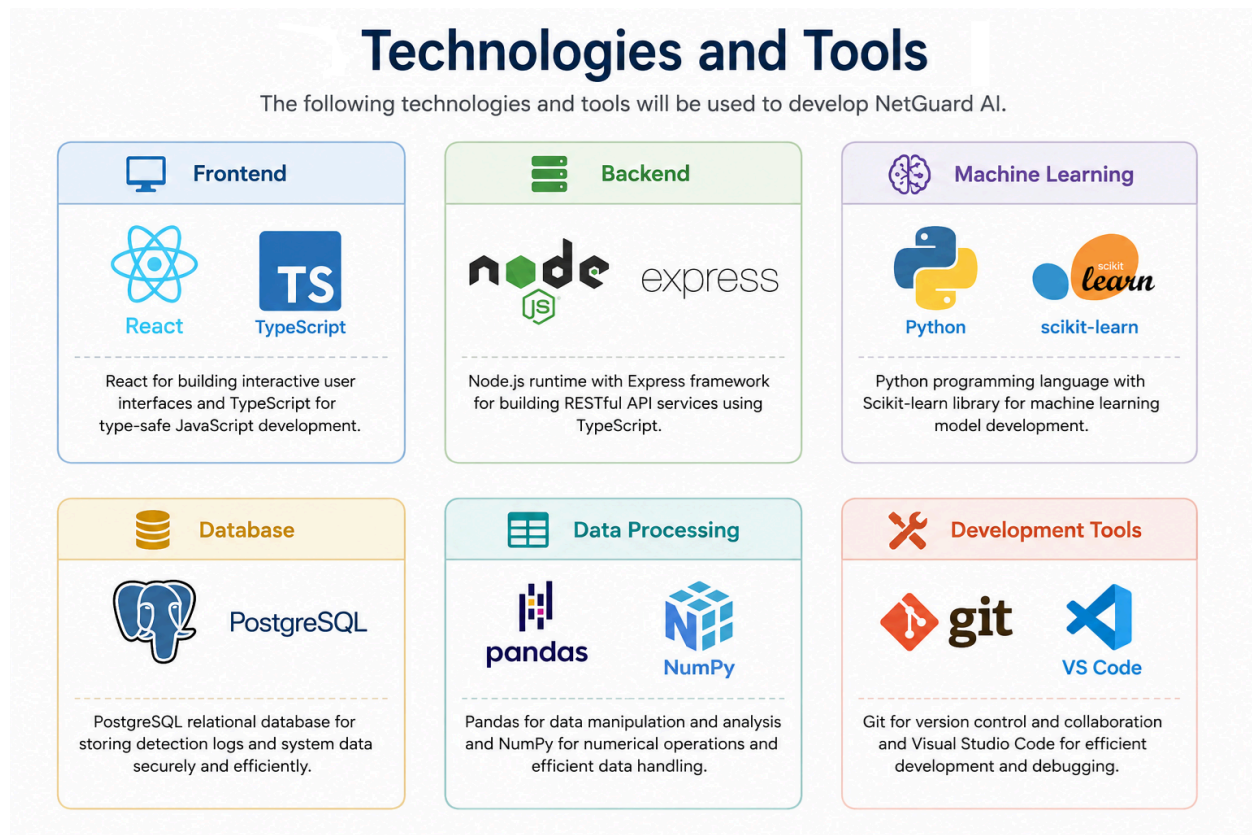
Phase 6: System Testing

The completed system will be tested using sample network traffic records to verify detection accuracy and system performance.

System Architecture



Technologies and Tools



Database Design

The system will maintain a traffic log repository containing:

- Source IP Address
- Destination IP Address
- Protocol Type
- Prediction Result
- Confidence Score
- Detection Timestamp

This information will support monitoring and historical analysis.

Expected Outcomes

The proposed system is expected to:

- Successfully classify network traffic as normal or suspicious.
- Demonstrate the effectiveness of machine learning in intrusion detection.
- Provide a simple and user-friendly monitoring dashboard.
- Store and retrieve traffic logs efficiently using PostgreSQL.

Evaluation Metrics

The machine learning model will be evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

The overall system will also be evaluated based on response time and usability.

Conclusion

The project aims to provide a practical and lightweight approach to network intrusion detection using machine learning techniques. By combining traffic monitoring, anomaly detection, and web-based visualization, the project demonstrates how artificial intelligence can enhance network security while remaining simple enough for deployment in educational and small-scale environments.